

Scaling in the Cloud

WHAT IS SCALING?

- Scaling is the characteristic of a system, model or function that describes its ability of growing or shrinking whenever required.
- In computing, scaling represents the capability of a system or application to deal with varying workload efficiently without bringing in a situation where resource shortage hampers performance or resource surplus increases the computation cost.
(or)
- In simple words, scaling is defined as the ability of being enlarged (or shrunk) for accommodating growth (or fall-off) to fulfill the business needs.
- A system or application architecture can be termed as scalable if its performance improves on adding new resources and the improvement is proportional to the capacity added

scalability

- The scalability of a scalable system is measured by the maximum workload it can competently handle at any particular moment.
- The point at which a system or application cannot handle additional workload efficiently, any more, is known as its limit of scalability.
- Scalability reaches its limit when a system's architecture does not support scaling anymore or some critical hardware resource run out.

SCALING IN TRADITIONAL COMPUTING

- In traditional computing environment, once the resource capacity of a system is enhanced manually the status of the system is retained till further human intervention, even if those resources do not get utilized.
- This under-utilization causes wastage of resources and increases the cost of computing.
- The main reason behind this problem is that the infrastructure architecture is not dynamic in traditional computing system, which prevents implementation of dynamic scaling.
- Static scaling requires system shut-down (system to restart) and hence is avoided unless it becomes extremely essential.
- For this reason, in traditional static scaling environment, although resource capacity expansion was sometime considered a possible option, capacity contraction was beyond imagination because service disruption had to be avoided.
- In contrast to the actual definition, conventionally, scalability has been about supplying additional capacity to a system.
- It was uncommon to reduce capacity of a system, although technically it was always possible. No one ever thought of migrating to a system of lesser capability, even when workload was reduced below the average level.
- System designers have built computing systems by arranging resources to meet peak demand, wasting resources and increasing cost.

SCALING IN CLOUD COMPUTING

- Scaling is one of the attractive attributes of cloud computing. Scaling in cloud is dynamic in nature and apart from a few special cases it is automatic too.
- In dynamic-automatic scaling, the system resource capacity can be altered while a system is running. Large pool of virtualized resources promises to adjust variable workload by allowing optimum resource utilization.
- The scalability in cloud computing is offered in a transparent manner and the automatic scaling ability gives the impression of infinite resources to its consumers.
- Many cloud service providers even claim to offer infinite scalability although in reality that is not true. Even the largest players may at some moment face a scalability problem if cloud computing usage rate increases abruptly, beyond anticipation.
- The scaling ability provided by cloud computing is particularly helpful for applications and services to meet unpredictable business demand.
- If overall resource requirements are planned properly in advance at the provider's end, then on-demand resource delivery nature of cloud computing model can make it robust and economic.

Scaling in Cloud is Reversible

- Virtualization has introduced the idea of dynamic infrastructure (using virtual computing resources) which has ignited dynamic resource provisioning.
- Dynamic scaling of computing system has become a reality through the dynamic resource provisioning approach where a system can be scaled without being compelled to restart.
- Thus, addition and removal of resource capacity to and from a running system has become viable and it has become much simpler to exploit extra capacity in cloud computing.
- Any time during the running of a system, extra resource capacity can be given back to the resource pool and can be claimed again whenever it is required.
- It should be noted that in any computing environment, the scaling-down is as important as scaling-up from business point of view.
- Cloud computing addresses both of these issues with equal priority. But downward scaling is more critical to implement than upward scaling.
- In downward scaling, the challenge is to maintain performance while releasing resources. The presence of downward scaling capability has made cloud computing more attractive.

FOUNDATION OF CLOUD SCALING

Scaling in cloud is dynamic and automatic. The auto-scaling facility provides enormous advantages, both to the service provider and the service consumers. They can both gain in terms of service availability (during peaks), cost and business. This auto scaling feature in cloud computing has been achieved based on three pillars:

- Resource virtualization: It eases resource management tasks and reduces the complexity of system development.
- Resource sharing: It allows the optimal resource utilization by sharing resources among multiple users as well as applications.
- Dynamic resource provisioning: It supplies (or reclaims) the resources on-demand, in response to the increase (or, decrease) in workload.

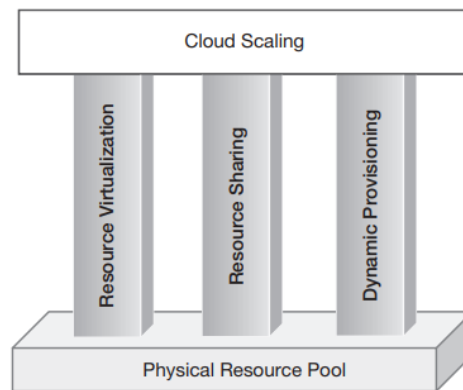


FIG 9.1: Foundations of cloud scaling

Scalability has to be kept in mind from the very beginning when sculpting the architecture of a system. A system's ability of scaling may depend on various parameters. Among them, the two important ones which are dealt meticulously in the development of scalable cloud computing system are:

- Capacity planning on regular basis and
- Load balancing

SCALABLE APPLICATION

- Application architecture should be well-suited to scale in a scalable architectural environment. While cloud provides ideal environment for scaling, applications must also be designed accordingly, having with scaling capability.
- Scalable computing system architecture lays the foundation for application scalability but cannot alone ensure scalability of application if the application is not suitably designed.
- On the other hand, a scalable application can only scale diligently with the varying traffic load if the system on which it is running is scalable. Hence, scaling of application depends on two layers which are, scalable application architecture and scalable system architecture.
- Cloud platforms provide rich auto-scaling capability to consumers. When used in combination with cloud-native applications, this capability makes the system highly effective.
- Sometimes, the enriched scaling characteristic of cloud computing may encourage lazy application developers to avoid the critical process of application development following scalable application architecture. But that can only result in low performance
- Application's scalability depends on its design, the types of algorithms applied and on the data structures used. Scalability of application also depends on the database system architecture.

SCALING STRATEGIES IN CLOUD

Cloud computing is empowered by its dynamic scaling capability. In the dynamic scaling approach, a system can be re-sized during its execution without restarting or interrupting any service. This critical task of dynamic capacity alteration can be done in two ways:

■ **Manually**: when a system can be scaled while running by executing appropriate commands through the application interface.

■ **Automatically**: when this type of scaling of the system can be implemented through programs that can automatically adjust system capacity by observing the actual demand.

The ability of manual capacity adjustment of a system during its operation is a huge task for any computing environment. But passing beyond this, the real power of cloud scaling lies in automatic scaling ability. Here, no human interaction is required. The system can adjust or resize itself on its own.

The dynamic auto-scaling is generally referred as auto-scaling which is also known as cloud scaling. Auto-scaling can be implemented in two different ways:

■ Scaling based on a predefined schedule known as proactive scaling.

■ Scaling based on current actual demand known as reactive scaling.

1. Proactive Scaling

- Application demand generally varies with time. Suppose, an e-commerce site is rarely accessed in the early morning or, an enterprise application gets the majority of hits during last two hours of the day. In such cases, where the expected increase or decrease of demand is known a pre-programmed plan is placed to automatically alter the resource capacity.
- Such scaling strategy that does not wait for workload to change, rather alters capacity in advance based on a pre-defined schedule is known as proactive scaling.
- These types of known situations of demand change can be categorized into two types. Separate planning is required to deal with those two types of situations.

Hence, the proactive scaling schedules are implemented in two different ways as

■ **Proactive cyclic scaling**: This type of proactive scaling event takes place at fixed regular intervals and by pre-defined times of the day, week, month or year. For example, an enterprise application may need to scale every business day from 11 AM to 3 PM which is their peak business hours.

■ **Proactive event-based scaling**: Major variations in traffic load may occur due to some scheduled business events like promotional campaigns or new product launch and else. For those cases, event-based proactive scaling is the best way out.

- Often through business and statistical analysis, the variable pattern of application or service demand becomes predictable.
- When using proactive scaling, the expected standard deviation should be considered. The exact requirements may not be possible to predict but the system should be ready for slight deviation from expectations. If for an application, 5,000 page views/ minute is expected during the peak hour(s) of a day and it is seen that the actual count is 5,050 it cannot be termed as unexpected. So, the provisioned resource capacity at any point of time should be able to handle the expected load with some room to spare. Understanding what comes under this 'room to spare' segment is a tricky part of capacity planning.

2. Reactive Scaling

- In this strategy, the system reacts immediately to changing demand of resources by adding or removing capacity on its own. Here, the decision is taken based on resource utilization.

- When utilization of processor or memory or some other resource reaches a certain threshold, more of that resource can be added into the environment by the system itself without any external intervention. Under this scaling technique, depending on a situation where the suitable parameters are identified at first to activate the auto-scaling process.
- System scales in response to the changing conditions of those parameters. This eliminates the need for any pre-scheduled action to handle scaling as it always remains unknown when those conditions may change.
- A scalable system should deal with varying workloads through proactive scaling approach as much as possible. Reactive scaling approach should be seen as a safeguard for absolutely unavoidable scenarios. Too much dependency on the reactive scaling strategy, without performing appropriate capacity planning to facilitate proactive scaling of a system may turn suicidal.

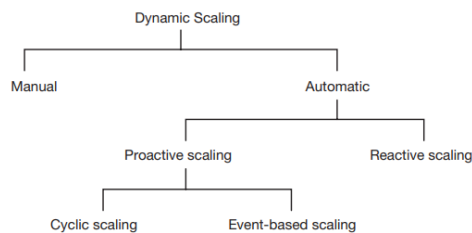


FIG 9.2: Classification of dynamic scaling approaches

- One vulnerability of reactive scaling strategy is that it may create trouble if not protected properly against attacks.
- An attack to computing system may cause malicious consumption of all resources including whatever newly been added. In such case, reactive scaling mechanism may fire a series of failed attempts to balance the gap by launching new resources while the cost increases sharply.

AUTO SCALING IN CLOUD

- In auto-scaling (also called as cloud scaling) mechanism, the system itself can increase the required resource capacity, automatically (and dynamically), when the demand of workload goes up.
- The mechanism can also release and return resources to the free pool when they are no more required. Auto-scaling facility helps to maximize resource utilization by automatically doing the work of scaling.
- The working of the auto-scaling mechanism in cloud has been briefly described in Figure 9.3. Auto-scaling allows scaling of computing resources both in the predictable and unpredictable circumstances.

Scaling in these two situations happens in following fashion:

- Unpredictably, based on specified conditions and
 - Predictably, according to defined schedule
- A set of conditions or rules can be specified based on application-specific metrics like resource utilization rates, number of simultaneous users or transactions per second etc.
 - For example, a pre-defined condition may say that additional (same) types of resource will be launched if the overall resource utilization rate reaches 90 percent and the last added resource will be released if overall resource utilization rate of the entire system falls below 45 percent.

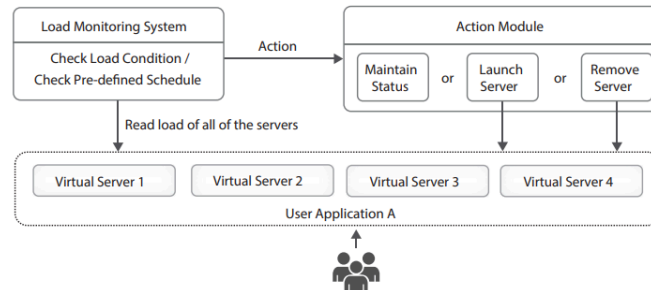


FIG 9.3: Working of the auto-scaling mechanism in cloud computing

- The monitoring and the checking modules of auto-scaling unit play vital roles in the autoscaling process.
- Monitoring module keeps on sending the load status to the checking module at regular intervals, while checking module decides the appropriate action depending on the current load status and stored pre-defined schedules.
- In the example shown in Figure 9.3, the load monitoring module launches an additional (same type of) server to support Application A, if all of the previously launched server(s) are loaded. In case of low utilization of all of the servers, the last added resource is revoked.

TYPES OF SCALING

A computing system can be scaled by providing additional hardware resources. There are two different ways of supplying and integrating these hardware resources into a system. Depending on this, the scaling approaches can be categorized into two types as **vertical scaling** and **horizontal scaling**.

Vertical Scaling or Scaling Up

One way of increasing resource capacity is to replace the existing component (processors, memory etc.) with a more powerful hardware component. Thus, the capacity can be increased (or decreased) by upgrading (or reducing) a resource quality to support raising (or falling) workload. This type of scaling where a resource component is powered up through replacement is called scaling up or vertical scaling. For example, a system with dual-processor capacity can be scaled vertically by replacing the processor with a quad-processor. In cloud computing, increase in system capacity through vertical scaling is implemented by provisioning and integrating additional hardware resources within an existing resource node, instead of introducing new node to handle increased demand. This is shown in Figure 9.4.

Advantages:

It is not a complex task to replace a component of a system since it does not require alteration in the system architecture.

vertical scaling approach has low management complexity and is less risky.

Disadvantages:

High power hardware components or specialized resources are expensive.

Capacity or power of any resource component always has an upper limit.

A sufficiently capable resource may not always be available or if available, the specialized hardware components may not be even affordable.

On the other hand, the hardware replacement may cause interruption to the service and thereby introduce a system downtime problem.

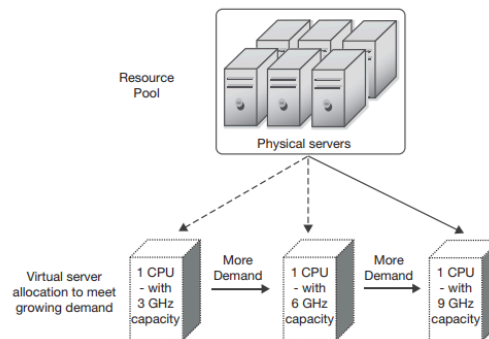


FIG 9.4: Computing resource scaled up by replacing it with more powerful resource as demand increases

Horizontal Scaling or Scaling Out

Resource capacity of a system can be increased by introducing additional resources (and without elimination of the existing). These new resources work together with the existing components. This type of scaling where the system is equipped with supplementary resources to expand the resource capacity is known as scaling out or horizontal scaling.

For example, a system with a dual-processor capacity can be scaled horizontally by adding another dual-processor.

cloud computing, the increase in system capacity through horizontal scaling is implemented by provisioning additional resource nodes instead of replacing resource components of existing nodes with powerful components. This is shown in Figure 9.5.

Advantages:

This approach does not involve any system downtime.

Specialized high-end hardware components are not required to scale out a system; generally available commodity hardware is used for horizontal scaling here.

This scaling approach is not limited by an individual component's capability as multiple similar type components (i.e., multiple processors or multiple memory units etc.) synchronize together in order to act as a powerful component.

Unlike vertical scaling, no resource components are thrown out of the system (through replacement).

Disadvantages:

This approach needs managing large number of distributed nodes which need to synchronize together, and hence involve complexity.

Presence of heterogeneous nodes further increases the complexity.

The other issue is that horizontal scaling is appropriate for running distributed software applications; hence existing applications developed to run in centralized environment are needed to be redesigned for a distributed computing environment.

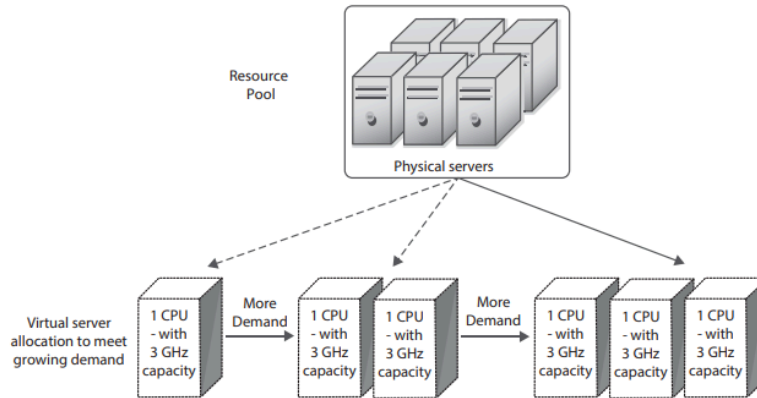


FIG 9.5: Computing resource scaled out by adding more number of same resources as demand increases

Horizontal scaling with homogeneous nodes is a significant simplification in the scaling approach. With homogeneous nodes, the associated activities for scaling like load balancing with round-robin scheduling and capacity planning becomes simple and easier.

Comparison Between Vertical and Horizontal Scaling

Table 9.1 Comparison between vertical and horizontal scaling

<i>Vertical Scaling</i>	<i>Horizontal Scaling</i>
It is known as Scaling up.	It is known as Scaling out.
This approach involves replacement of components or resource nodes.	This approach involves introduction of additional components or resource nodes.
It can be implemented in any type of computing environment.	It can only be implemented in distributed computing environment.
It has less management complexity.	The management of larger numbers of nodes increases system complexity.
It may cause service interruption as system needs to restart after replacing component.	It does not cause service interruption. No system-restart is required.
It has less influence on application architecture.	This approach has a more fundamental influence on application architecture.
A vertically scalable application tends to run on high-end hardware.	A horizontally scalable application tends to run on low-end (commodity) hardware.
Here, the loads are concentrated.	It spreads the load.
It requires the specialized hardware components.	It can be done using normal commodity hardware components.
Upgrading the capacity of resource beyond a certain level can become very expensive in this type of scaling.	It is always less expensive and the expense is always proportional to resource capacity.
The expansion is limited by any resource's maximum capacity.	The expansion does not depend on available capacity of hardware components.
It is not a long term solution for scaling.	It provides a long term solution for scaling.

The Appropriate Choice:

Although the architectural structures of the vertical and horizontal scaling approaches are different from one another, there are always trade-offs between the two models of scaling. The question raised here is whether it would be better to increase the power of the individual nodes or to combine the power of multiple nodes by distributing the load among them. The answer is not one. It differs on a case by case basis.

The decision depends on two aspects: **the computational and business requirements**.

computational viewpoint	business requirements
The choice is primarily linked with factors like the architecture of the system and design architecture of application	The availability of requirement, cost of implementation and system performance are the decisive factors. In some cases, vertical scaling fits more appropriate

HORIZONTAL SCALING IS MORE CLOUD-NATIVE APPROACH

Vertical scaling has limitations

It can only grow as per available resource components. Thus, the infinite scalability promise cannot be fulfilled with vertical scaling approach in the cloud computing environment where application load may reach the zenith on different occasions.

Horizontal scaling on the other hand can spread the load across multiple resource nodes and can thus support load unrestricted by any resource component's ability. The more resource nodes are added, the more load can be supported and thus this scaling approach is rather pertinent to deliver infinite scalability.

The other issue is that in an auto-scaling environment, replacing a component with another is not a suitable choice. Rather, adding and releasing resources to scale is much easier approach. Thus, the horizontal scaling approach becomes more cloud-native

Infinite scalability in cloud computing can only be achieved through the horizontal scaling approach.

PERFORMANCE AND SCALABILITY:

Applications are often measured by two important parameters like performance and scalability

Performance is what an individual user experiences but scalability is the number of users who happen to experience that performance.

Application performance depends on different parameters. For any application, **the response time** is considered the most important performance measure. Apart from this, **the elapsed time** for completing all of the activities determines the overall performance of any application.

Scalability refers to the number of concurrent users who have a positive (satisfactory) experience. If the application can maintain its performance with growing number of concurrent users then it is said that it scales well.

Suppose, with 1000 concurrent users the average response time of an application is 1 second but as the number of concurrent users grows to 2000, the average response time becomes 2 seconds; then the application is said to be not scaling well.

In an ideal scalable system, the response time should not increase with the number of the concurrent users.

An application may scale well without performing well. This may happen when the application handles many concurrent users with same performance but that performance is poor irrespective of the number of concurrent users.

Again, the performance tends to degrade with growing number of concurrent users. So there is always a clash between performance and scalability. This problem is settled by setting a threshold value which indicates the limit of scalability. The limit is determined by the maximum number of concurrent users for whom an application may perform consistently. Beyond this limit, the performance would degrade.

THE RESOURCE CONTENTION PROBLEM

scalability often faces problems due to resource contention.

In computing, resource contention is about the controversy over access to sufficient volume of resources. For example, when a network communication path cannot handle any more traffic towards a web application, the application suffers with scalability issue.

In this case, the resource bottleneck is the **network connection**.

Thus, generally it is not the number of concurrent users but the resource bottleneck problem that limits the sequence of scalability. Different applications have different stress points. System designers must therefore analyze the expected system load patterns and carry on tests to check how different scenarios create different stress points in the infrastructure.

Here for cloud based applications, the most likely initial stress points have been mentioned below:

- Process speed of application server
- Memory capacity and speed of application server
- Speed of the disk I/O operation of database server
- Network bandwidth

Increase in stressed resource's capacity is the only way to scale a system beyond some bottleneck or stress point. But the fact remains that elimination of one bottleneck only reveals another one.

Hence, the system components should be synchronized to push any such introductions of bottleneck problems as far away as possible.

CLOUD BURSTING: A SCENARIO OF FLEXIBLE SCALING

When dynamic provisioning approach is applied to provision resources from an external cloud system to support some overloaded application running on organization's internal computing environment, then this state of affairs is referred to as **cloud bursting (or cloud peering)**.

In cloud bursting architecture, initially the resources of organization's internal computing system are consumed by applications.

With growing resource demand whenever a pre-defined capacity threshold is reached, the dynamic resource provisioning mechanism 'bursts out' and starts consuming essential resources from a cloud, predominantly a public cloud.

When the internal computing set-up is a private cloud facility, in such cases, the deployment becomes a hybrid application.

When the existing private cloud infrastructure cannot handle the load spikes, the workload is migrated to some public cloud deployment. Again when the load of application reduces, the public-cloud-based computing resources are released and the system gets back to the on-premises environment.

Thus, the workload shifts between external and internal hosting to manage the changing resource demands without any knowledge of the users

The foundation of the cloud bursting architectural model is based on two functionalities like **automated scaling listener and resource replicator**. The automated scaling listener decides when to redirect load to an external cloud system. Resource replication maintains the system status during load switch between on-premises system and cloud environment. Figure 9.6 represents the architectural model of cloud bursting

As shown in Figure 9.6, the automated scaling listener monitors the load of on-premises service 'A'. When on-premises service load exceeds the threshold value, the scaling listener redirects traffics towards the replicated implementation of service 'A' in the external cloud. In the meantime, the resource replicator copies the current state and data of on-premises service 'A' to external cloud in order to keep the systems synchronized. Service 'A' remains redundantly pre-deployed in the external cloud in an inactive mode until cloud bursting occurs.

In Figure 9.6, when the on-premises service 'A' becomes fully-loaded with traffics from service consumers 1 and 2, the automated scaling listener redirects the traffics from service consumer 3 towards the external cloud implementation of Service 'A'. When load decreases, the external cloud service is released and the system retreats to the internal on-premises system

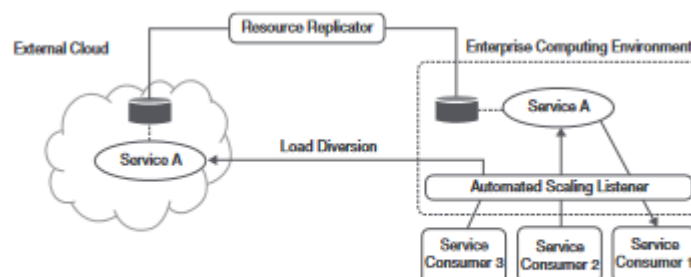


FIG 9.6: Cloud bursting architecture

Growth is a basic goal of any business. But infrastructural support is one primary necessity to facilitate this growth. Hence, apart from flexibility of operation, what any business looks for is scalability. Business scalability largely depends on the scalability of the computing system supporting the business. Business applications must be able to support growing workload by processing sudden traffic escalation efficiently.

Scalability is about customer satisfaction as it can process growing extent of incoming traffic with consistent performance. More customer satisfaction means more business. Studies show that a few milliseconds of delay in page load time reduce business by significant percentage.

Amazon observed 1 percent decrease in retail revenue caused by 100-millisecond delay. Analysis by Google showed 20 percent decrease in traffic due to an additional 500-millisecond delay in page response time.

Hence, scalability ultimately becomes a vital business concern.

In cloud computing, businesses can take advantage of nearly infinite scalability by using the right scale to deploy their applications. The automatic scaling facility to grow during traffic demands and shrink during less workload allows businesses to focus on their core business objectives.